Effect of Remote Work on the Child Penalty: Evidence from the United States

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May 14, 2026

Most recent draft here.

Abstract

We study the effect of the post-COVID expansion of remote work on the child penalty. Our empirical design builds on three steps: using pseudo-panels to estimate child penalties across occupations, exploiting cross-occupational variation in remote work adoption, and using synthetic controls to adjust for differential pre-COVID trajectories. Remote work reduces the hours penalty for mothers: each percentage point increase in an occupation's remote work share raises mothers' hours worked by 0.23%, eliminating one-sixth of the baseline motherhood penalty for the average remote occupation. We find no effects on employment, income, or wages for mothers, and no average effects on men.

Keywords: Working from Home, Child Penalty, Gender Inequality

JEL codes: J13, J16, J22, J31

1 Introduction

Since the 1970s, the gender gap in earnings and employment in the United States has narrowed substantially (Goldin, 2024). Nevertheless, significant disparities persist: women participate less in the labor force than men and, when employed, earn lower wages. A growing literature shows that these gendered disparities are largely explained by the child penalty: the differential impact of parenthood on labor market outcomes between women and men (Kleven et al., 2019, 2024a; Cortés and Pan, 2023). Further progress in gender equality likely requires changes in child penalties, yet traditional family policies, including parental leave, childcare subsidies, and flexible scheduling, have had limited effects, even in Scandinavia (Kleven et al., 2024b; Olivetti and Petrongolo, 2017). We study whether one of the largest natural experiments in labor markets in recent decades, the expansion of remote work after the COVID-19 pandemic, under which roughly 30% of US workdays now occur remotely (Aksoy et al., 2025), has affected the child penalty in the United States.

Theory alone cannot sign the impact of remote work on child penalties. On one hand, remote work provides flexibility and reduces commuting time, which can help parents, and especially mothers, who tend to bear a disproportionate share of caregiving responsibilities, balance work and family (Goldin, 2014; Goldin and Katz, 2016). Mothers who would otherwise reduce their hours after childbirth may find it easier to maintain their pre-birth schedule, which would lower the child penalty. On the other hand, remote positions may carry an amenity premium: workers value flexibility and accept lower wages to obtain it (Mas and Pallais, 2017). If mothers disproportionately select into remote work for its non-wage benefits, the result could be lower earnings and reduced career progression (Bloom et al., 2015). A third possibility is that working from home while caring for young children lowers hourly productivity, so that even if mothers work more hours, their output and earnings do not increase proportionally. Experimental evidence supports this channel: Ho et al. (2025) find that women randomly assigned to work from home rather than an office complete tasks 18% more slowly, with the productivity loss driven by fragmented work sessions as domestic responsibilities interrupt sustained focus. These forces oppose each other, making the net impact of remote work on the child penalty an empirical question.

Three distinct challenges arise in studying this question. First, the standard event-study approach to estimating child penalties requires panel data, which are limited in size. The PSID and NLSY track only a few thousand parents; estimating child penalties separately for each of 94 occupations requires orders of magnitude more observations. Second, while public datasets like the ACS can measure current remote work arrangements, precise estimates of the treatment require tracking remote work incidence *before* COVID across occupations, information that standard surveys do not provide. Third, the adoption of remote work is not random across occupations. Remote-friendly occupations, disproportionately high-skill, white-collar jobs, were already on differential child penalty trajectories before the pandemic, making the standard parallel trends assumption unlikely to hold.[1] This is a first-order concern: as we document, the employment penalty in eventually-remote occupations was trending downward relative to non-remote occupations throughout 2001–2019, well before remote work expanded.

We address these challenges in three steps. First, we use the pseudo-panel approach of Kleven (2025), which matches parents to observationally similar non-parents in the ACS ($\sim$3 million observations per year), providing sufficient statistical power for occupation-level estimation. We exclude the 2020 and 2021 ACS to avoid bias from pandemic-specific disruptions unrelated to remote work (school closures, lockdowns, labor hoarding). Second, we use Lightcast (formerly Burning Glass Technologies), which classifies the universe of US online job postings as remote or on-site since 2015, to measure the change in remote work share for each occupation before and after COVID. Third, we apply synthetic control difference-in-differences (SC-DiD) to construct counterfactual trends for each occupation, adjusting for the differential pre-COVID trajectories that would bias standard DiD.[2]

We find that remote work reduces the hours child penalty for mothers but does not affect employment, income, or wages. Each additional percentage point of remote work raises mothers' log hours by 0.23% ($p = 0.029$), implying approximately 2% more hours for the average remote occupation, which experienced an 8.9 pp increase in remote work share. This effect eliminates roughly one-sixth of the baseline hours penalty for mothers in remote occupations, comparable in magnitude to the 22% reduction from Australia's Fair Work Act (Ciasullo and Uccioli, 2023). The

[1] Possible explanations include the diffusion of communication technologies (video conferencing, cloud collaboration tools) that disproportionately benefitted knowledge-work occupations, making them incrementally more compatible with flexible work arrangements even before remote work expanded broadly.

[2] Our algorithm is a special case of the Synthetic IV algorithm of Gulek and Vives-i Bastida (2024) where the instrument is the treatment. We perform their algorithm instead of the more commonly used Synthetic-DiD algorithm of Arkhangelsky et al. (2021) since the former works also with continuous treatments while the latter requires dummy treatments and never-treated units.

hours effect operates broadly across the distribution, shifting mothers from low part-time hours toward substantial part-time or full-time schedules. We find no average effects on men. Despite the hours gains, we find no significant increase in income or wages for mothers, and the employment point estimate is negative though imprecisely estimated. Evidence from the American Time Use Survey points to a mechanism: mothers in remote occupations sharply increased the time spent supervising children while working after COVID, consistent with lower hourly productivity offsetting the hours gains.

The hours result is robust across several specifications varying the child-age bins, the matching moments, and the weighting scheme. We backtest the synthetic control weights by restricting the training period to 2001–2015 and assigning a placebo treatment to 2016–2019; all placebo tests yield precise nulls. We validate results using the rotating-panel approach of Gulek (2024) applied to the CPS; the CPS estimate is too imprecise to independently confirm the hours effect but is consistent in sign and magnitude. We find no evidence of anticipatory occupational sorting: non-parent employment rates do not differentially improve in remote occupations after COVID.

We contribute to two literatures. First, we contribute to research on gender inequality and child penalties. Child penalties are notoriously unresponsive to government policies: expansions of parental leave, childcare subsidies, and gender quotas have had limited effects on the earnings gap between mothers and fathers (Kleven et al., 2024b; Olivetti and Petrongolo, 2017). A notable exception is Ciasullo and Uccioli (2023), who provide the first causal evidence that work arrangements affect child penalties: exploiting Australia's 2009 Fair Work Act, which entitled mothers to request reduced hours on predictable, permanent contracts, they find an 8 pp increase in labor force participation and a 22% reduction in the hours penalty. We contribute to this literature by showing that remote work, a market-driven change in work arrangements rather than a government mandate, also reduces the hours child penalty in the United States.

Second, we contribute to the literature on remote work and labor markets (Bloom et al., 2015, 2022; Aksoy et al., 2025; Emanuel and Harrington, 2024). While much of this literature studies productivity, wages, or worker preferences, we provide occupation-level estimates of how remote work affects child penalties using synthetic controls to address pre-existing differential trends. Most closely related, three concurrent papers study the relationship between remote work and child penalties.

Harrington and Kahn (2026) use the American Time Use Survey to measure changes in the share of work hours performed remotely across college-degree fields during 2009–2019, and correlate these with changes in the employment gap between mothers and non-mothers in the ACS. They find that a 1 pp increase in the share of workers working primarily from home is associated with a 1.3 pp narrowing of the mother–non-mother employment gap. Our contribution relative to Harrington and Kahn is threefold. First, we estimate causal child penalties from event-study designs, not the raw gap between mothers and non-mothers, which confounds the causal effect of children with selection into motherhood. Second, we document that eventually-remote occupations were already on differential child penalty trajectories before the pandemic, a first-order threat to any difference-in-differences comparison of remote and non-remote groups. Adjusting for these pre-trends via synthetic controls, we find that remote work does not reduce the employment penalty. Third, we show that the causal effect of remote work operates on the intensive margin: remote work reduces the hours penalty conditional on employment, a result not documented previously.

Two papers investigate the effect of remote work on child penalties in other settings. Scott and Sundberg (2025) use Swedish administrative data and a triple-difference design, comparing child penalties before and after the pandemic across occupational quartiles of remote work exposure. They find that remote work failed to reduce motherhood penalties across any margin, including hours, and develop a household model in which the productivity penalty from remote work and increased returns to home production can offset the time savings from eliminated commuting, producing null aggregate effects. In contrast, Zarate (2025) studies Argentina, Brazil, Chile, and Colombia using pseudo-panels, finding that remote work reduces the employment child penalty by about 4 pp relative to a baseline of 17 pp. Our contribution to this growing body of work is threefold: we document the differential trajectories of child penalties in remote and non-remote occupations, we show how synthetic controls can address these pre-trends even in continuous treatment environments, and we document that mothers increased childcare while working in remote occupations, which may explain the absence of income and employment gains. Future work can replicate the synthetic control approach developed here across the many countries where large cross-sectional or rotating-panel datasets exist (Kleven et al., 2024a; Donovan et al., 2023).

2 Data

2.1 Remote work

We use Lightcast (formerly Burning Glass Technologies), which covers nearly all US online job postings (∼40 million per year) (Hansen et al., 2023). Lightcast classifies each posting as remote or on-site based on text analysis of the job description. Unlike worker-reported surveys, this measures the *demand* side of remote work: the share of positions that employers advertise as remote. This captures employer willingness to offer remote arrangements, though the gap between advertised and realized remote work may vary across occupations. We aggregate to 94 three-digit SOC occupations and compute the remote work share in 2017–2019 (baseline) and 2023–2025 (post).[3] Our treatment variable is ΔR_o, the percentage-point change in remote work share for occupation o between these two periods. We also classify occupations as remote or non-remote based on whether ΔR_o is above or below the median.[4]

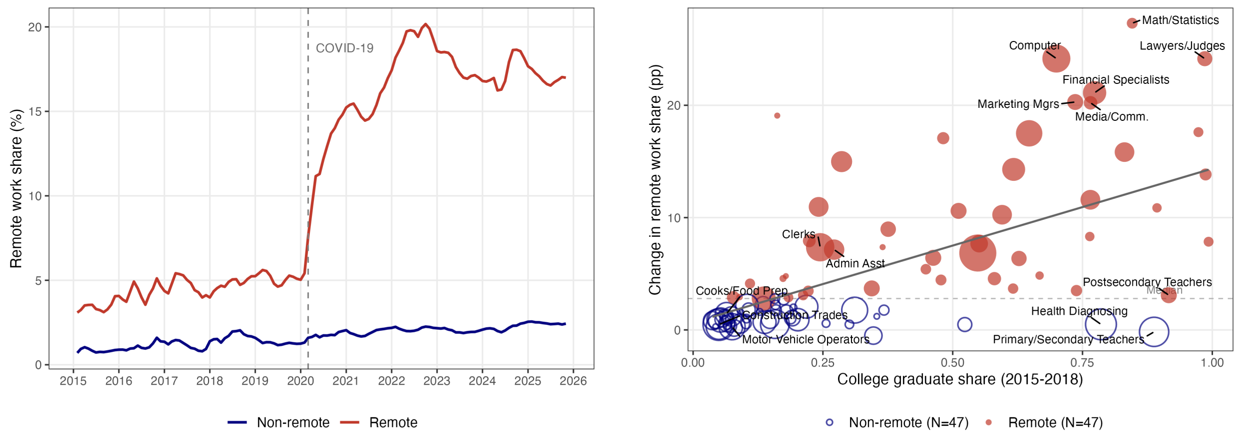
Figure 1 documents the variation in remote work adoption that identifies our results. Panel (a) plots the monthly remote work share for remote and non-remote occupations from 2015 through 2025. Before COVID, both groups were essentially flat at low levels. After March 2020, remote occupations experienced a sharp, persistent increase while non-remote occupations barely moved. The pre-COVID stability confirms that the pandemic was the primary driver of remote work adoption, rather than a gradual trend that happened to accelerate. Panel (b) shows the cross-sectional variation underlying this aggregate pattern. Remote work adoption is positively correlated with skill intensity: low-skill occupations saw minimal increases, while high-skill occupations like computer, legal, and marketing experienced increases exceeding 20 pp. Two features of the treatment variation are worth noting: there is large heterogeneity across occupations in employer-offered remote work, and remoteness is not randomly assigned; remote and non-remote occupations differ

[3] We exclude 2020 from all analyses to avoid contamination from the acute phase of the pandemic. In the ACS, the post period begins in 2022 (excluding 2020–2021). In the CPS, the post period begins in 2021. The difference arises because ACS employment questions refer to the prior year, while CPS questions refer to the past week. Note that ACS "usual hours worked per week" refers to a typical week at the time of the survey, while employment and income refer to the prior year. Excluding the 2021 ACS is therefore conservative for the hours variable, which captures 2021 conditions rather than 2020, but we do so to maintain a consistent sample across outcomes.

[4] Lightcast measures *posted* remote work availability, not realized remote work arrangements. If measurement error in ΔR_o is classical (uncorrelated with the true remote share and with child penalty changes), it attenuates the continuous estimates, implying that the true effect per percentage point of actual remote work could be larger than we report. If the gap between posted and realized remote work varies systematically across occupations, the bias could go in either direction.

systematically in skill intensity and likely in other unobserved characteristics correlated with child penalty trends.

Figure 1: Remote Work Adoption by Occupation



(a) Remote work share over time

(b) Occupation skill and remote work adoption

Panel (a): monthly share of job postings offering remote work, 3-month moving average. Red: remote occupations ($\Delta R_o \geq$ median); blue: non-remote. 2015–2018 from Lightcast extract, level-adjusted to match at 2019; 2019–2025 from Hansen et al. (2023). Dashed line: March 2020. Panel (b): each bubble is a 3-digit SOC occupation, sized by employment. x-axis: college graduate share (ACS, 2015–2018). y-axis: ΔR_o, change in remote work share, 2017–19 vs. 2023–25. Dashed line: median $\Delta R_o = 3.0$ pp. Red: remote ($N = 47$); blue: non-remote ($N = 47$). Line: WLS fit.

2.2 ACS

Our primary dataset is the ACS, 2001–2024 (∼3 million observations per year). We restrict the sample to individuals aged 25–55 who had their first child between ages 25 and 45. Following Kleven (2025), we construct pseudo-panels by matching parents to observationally similar non-parents based on age, education, race, and state. Occupations are assigned at the time of observation. We measure four outcomes: employment last year, log usual hours, log earnings, and log hourly wages. We partition the pre-COVID years into four five-year periods (2001–05, 2006–10, 2011–15, and 2016–19) and define 2022–24 as the post period, with 2016–19 as the reference. The five-year bins provide enough observations for precise occupation-level estimation, and the four pre-treatment periods enable us to document child penalty trends over time.

2.3 CPS

We complement the ACS with the CPS monthly files, 2000–2025 (excluding all individuals whose 4-8-4 rotation group overlaps with calendar year 2020, which drops all 2020 interviews as well as some late-2019 and early-2021 interviews from individuals straddling the pandemic onset). The CPS rotating panel structure (4 months in, 8 out, 4 in) allows observing individuals before and after their first child's birth. Following Gulek (2024), individuals who have no child in their first interview cycle but have a newborn in their second cycle are assigned event time $t = -1$, providing a clean pre-birth observation without matching. Since the post-COVID window is short and occupation-level estimation requires statistical power, we rely on the ACS for our main results and use the CPS as a robustness check. Standard errors in all CPS analyses are clustered at the individual level to account for serial correlation across interview rounds within the rotating panel.[5]

2.4 ATUS

We use the American Time Use Survey (ATUS), 2003–2024 (excluding 2020–21), to understand how remote work may differentially affect parents' productivity across genders. The ATUS is a one-day time diary administered to a subsample of CPS respondents (∼10,000 per year). It records *secondary childcare*: minutes during which a respondent supervises an own child while engaged in another activity. We focus on secondary childcare that occurs during paid work activities. Occupations are classified as remote or non-remote using the same binary median split as in the main analysis.

3 Empirical Strategy

Our empirical design has three layers: we first show how we estimate child penalties across occupations and time periods, then document pre-trends in child penalties across remote and non-remote occupations, and finally apply a SC-DiD algorithm to adjust for these pre-trends and identify the causal effect of remote work intensity on child penalties.

[5] Clustering at the individual level is appropriate whenever the same person is observed multiple times, as in the CPS rotating panel, PSID and NLSY panels, although this serial correlation is often ignored in the literature.

3.1 Estimating child penalties

The event-study approach to estimating child penalties uses panel data on men and women who become parents. For each occupation o and gender g, we estimate:

$$\begin{aligned} Y_{iot}^{g} &= \sum_{j\neq -1} \alpha_{o,j}^{g}\, \Delta D_{i,t-j} + \theta_{o}^{g} W_{it} + \epsilon_{iot}^{g}, \\ \ln(Y_{it}^{g,o}) &= \sum_{j\neq -1} \alpha_{o,j}^{g}\, \Delta D_{i,t-j} + \theta^{g,o} W_{it} + \epsilon_{it}^{g,o}, \end{aligned} \tag{1}$$

where $\Delta D_{i,t} = 1$ if individual i had their first child at time t, and j indexes event time relative to first birth, with $j = -1$ as the omitted category. The first equation estimates the effect of parenthood on employment, where the sample includes all individuals in occupation o and gender g. The second equation estimates the effect on conditional outcomes (log hours, log income, log wages), where the sample is restricted to employed individuals in occupation o; hence occupation appears as a superscript denoting a sample selection rather than a subscript. The controls W_{it} include age and calendar year fixed effects, as is standard in this literature (Kleven et al., 2019, 2024a), and also education fixed effects following Cortés and Pan (2023), to account for the fact that education is correlated with both labor market outcomes and the timing of fertility. Education controls are particularly important in our setting: if the pandemic changed the timing of parenthood differentially for high- and low-skill workers, omitting education would amplify the bias for precisely the occupations where remote work expanded most.

This design historically required panel data, not because it exploits within-individual variation (the specification does not include individual fixed effects), but to observe eventual parents before they become parents. Since available panel datasets are often too small for detailed heterogeneity analysis (the PSID and NLSY provide only a few thousand parents each), two methods have been proposed to overcome this limitation. Kleven (2025) developed the *pseudo-panel* approach, which uses exact matching to construct counterfactual non-parents from large cross-sections. Building on this work, Gulek (2024) showed that rotating panels can estimate (1) directly, because the regression requires only one pre-birth observation ($t = -1$) per individual, exactly what the CPS rotating panel provides. Since our setting requires statistical power for occupation-level estimation with a short post-treatment window, we rely on the ACS pseudo-panel for our main results and use the CPS rotating panel as a robustness check.

We implement the pseudo-panel as follows. In ACS year t, we observe parents whose eldest child is age $a \in \{0, 1, \dots, 10\}$. Since our main outcomes refer to the prior year (when children were approximately one year younger), we assign event time $j = a - 1$ for parents with children aged $a \geq 1$. We do not use this method to assign $t = -1$ to parents with children aged 0, because the imprecise timing of birth within the survey year would contaminate the pre-birth observation. Instead, following Kleven (2025), we match parents to individuals without children observed in the prior ACS year, who are one year younger but share the same education, race, and state, and assign them $t = -1$. We match only on $k = 1$ (one year prior) rather than $k \in \{1, \dots, 5\}$ as in Kleven (2025). This is sufficient given the established absence of pre-trends in child penalty estimation, and it avoids contamination from the 2020 and 2021 ACS years, which we exclude from the analysis.[6]

For employment outcomes, we convert level estimates to percentage child penalties following Kleven et al. (2019): $\alpha^{\%} = \alpha/(\bar{Y}_{\text{parents}} - \alpha)$, where $\alpha < 0$ denotes a penalty and $\bar{Y}_{\text{parents}}$ is the mean outcome for parents. Log outcomes are already in percentage terms.

When estimating child penalties for each of 94 occupations separately, small cell sizes, particularly for parents with older children, make the fully dynamic specification in (1) imprecise. We therefore also estimate a static version that replaces the event-time dummies with a single parenthood indicator D_{it}, yielding one precisely estimated parameter per occupation-period cell. The pre-trend analysis in Section 3.2 uses this static specification. We return to heterogeneity by child age in the SC-DiD stage, where synthetic control weights are computed jointly across child-age bins (Section 3.3).

3.2 Child penalty trajectories across remote and non-remote occupations

To document how child penalties have evolved over time across remote and non-remote occupations, we estimate a version of (1) that separates the baseline child penalty from its period-specific deviations. For each occupation group o and gender g:

$$Y^g_{iot} = \gamma^g_o D_{it} + \sum_{\tau \neq \tau_0} \alpha^g_{o,\tau} \left(D_{it} \times \mathbb{1}\{t \in \tau\} \right) + \theta^g_o W_{it} + \epsilon^g_{iot}, \tag{2}$$

[6] For example, to assign $t = -1$ for ACS 2023 parents, we draw matched non-parents from ACS 2022 only. Matching on $k \in \{1, \dots, 5\}$ would require using ACS 2018–2022, pulling in the excluded COVID years.

where γ_o^g captures the baseline child penalty for gender g and occupation group o that is common across time, and $\alpha_{o,\tau}^g$ captures the differential child penalty in period τ relative to the reference period τ_0 (2016–19). An analogous specification with $\ln(Y_{it}^{g,o})$ is estimated for conditional outcomes, as in (1). We estimate (2) separately for remote ($\Delta R_o \geq$ median) and non-remote ($\Delta R_o <$ median) occupations.

Figure 2 plots the estimated $\alpha_{o,\tau}^g$ coefficients. Panel (a) shows that women's employment penalty followed strikingly different trajectories in remote and non-remote occupations throughout 2001–2019: the penalty in remote occupations was declining over time relative to non-remote occupations, with a larger penalty in the early 2000s that gradually decreased toward the 2016–19 baseline. Panel (b) shows that men exhibit no visually apparent differential patterns.[7] These differential pre-trends in women's employment penalties mean that a simple comparison of remote and non-remote occupations before and after COVID would confound the effect of remote work with the pre-existing trajectory. These trends predate the pandemic, but we do not know how they would have evolved absent COVID. Parametric adjustments such as linear trends are unappealing, as the pandemic is a sufficiently large shock that it could alter the shape of the trend itself. To make progress, we rely on synthetic controls, which construct a data-driven counterfactual for each occupation's child penalty trajectory.

3.3 SC-DiD

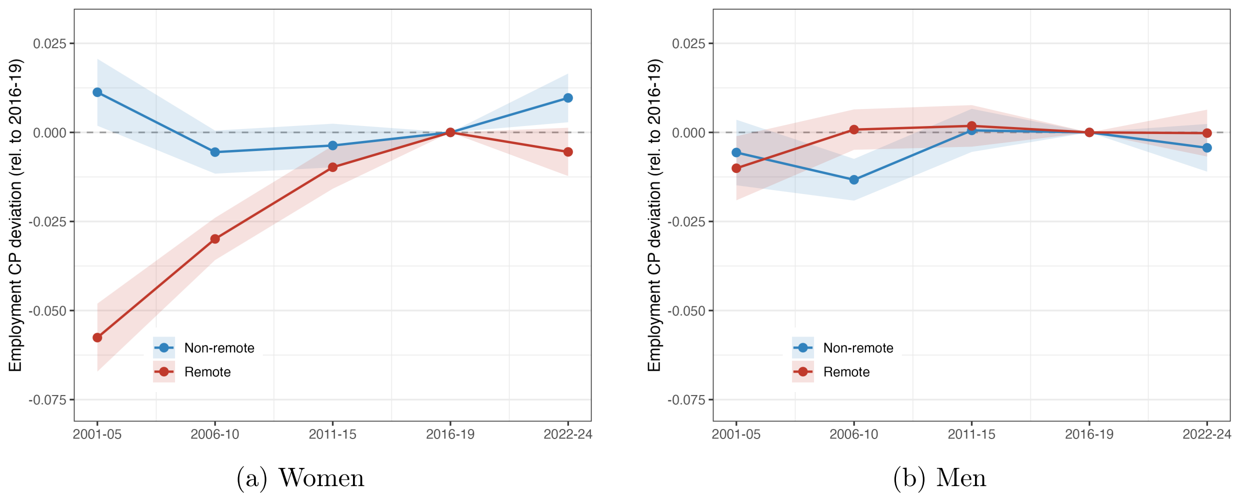
We model the occupation-level child penalty as:

$$\alpha_{o,\tau}^g = \beta^g \Delta R_o \cdot \mathbb{1}\{\tau = \tau_1\} + f_o + f_\tau + \lambda_o' \theta_\tau + \epsilon_{o,\tau}, \tag{3}$$

where ΔR_o is the change in remote work share induced by the pandemic, f_o and f_τ are occupation and time-period fixed effects, and $\lambda_o' \theta_\tau$ captures unobserved confounders generating differential trends. If we could observe these confounders, we would control for them directly and eliminate the omitted variable bias. Since we do not know the exact reason behind these trajectories, we use synthetic controls to indirectly proxy for these confounders. Identification requires that the interactive fixed effects $\lambda_o' \theta_\tau$ are low-rank relative to the number of pre-periods and matching

[7] Appendix Figure A.1 shows all four outcomes by gender; pre-trends are parallel for conditional outcomes (log income, log hours, log wages).

Figure 2: Child Penalty Trajectories: Remote vs. Non-Remote (ACS, Employment)



Employment child penalty deviations $\alpha^g_{o,\tau}$ from (2), expressed relative to the reference period (2016–2019, normalized to zero). Red: remote occupations ($\Delta R_o \geq$ median). Blue: non-remote. ACS pseudo-panel, robust SEs. Women (left): remote occupations show a differential downward trend before COVID. Men (right): no differential pattern. Appendix Figure A.1 shows all four outcomes by gender.

moments, so that the synthetic control weights can recover the unobserved loadings from the observed pre-treatment trajectories. Put differently, we require that the residual variation in remote work adoption between occupations and their synthetic controls is orthogonal to occupation-specific shocks affecting child penalties in the post period.

The empirical challenge is that ΔR_o is not a binary treatment: it is a continuous treatment, with some occupations far more exposed to remote work than others. For example, both administrative assistants and computer-related occupations observed increases in remoteness, but the latter was three times as exposed as the former, and we want to exploit this variation for identification. If we were working with a binary treatment, we could use the Synthetic DiD approach of Arkhangelsky et al. (2021). Instead, to incorporate the continuous treatment, we use a version of the Synthetic IV algorithm of Gulek and Vives-i Bastida (2024). The SIV algorithm converges to what we show below when the instrument equals the treatment; that is, we assume no endogeneity in the treatment and are concerned only with omitted variable bias from unobserved confounders. When the instrument equals the treatment ($Z_o = \Delta R_o$), the SIV first stage becomes the identity and the algorithm reduces to: (i) construct synthetic controls matching pre-treatment outcome trajectories, (ii) debias both outcomes and treatment using the same weights, (iii) regress the debiased outcome change on the debiased treatment.

Intuitively, for each occupation in our sample, we want to create a synthetic occupation that would have followed the same child penalty trajectory absent the remote work shock. To do this, for each gender, we find synthetic control weights that best match the pre-COVID trends in child penalties jointly across employment and hours outcomes. Matching jointly across outcomes improves the signal-to-noise ratio when outcomes are driven by related confounders (Sun et al., 2025), enables an apples-to-apples comparison across outcomes (the same weights are used for all outcomes, so differences in treatment effects across outcomes cannot be driven by differences in synthetic controls), and provides an additional diagnostic: SC-DiD should not create pre-trends in untargeted outcomes. The hours coefficient is similar under hours-only separate weights (+0.0019, $p = 0.042$), confirming that the result does not depend on joint matching. Since we have many donor units and few pre-periods, we further enforce the weights to match trajectories across four child-age groups: age 0, ages 1–2, 3–5, and 6–10.

Denote these four age groups by $j \in \{j_1, j_2, j_3, j_4\}$ and the five-year time periods by $\tau \in \{\tau_{-3}, \tau_{-2}, \tau_{-1}, \tau_0, \tau_1\}$. The algorithm proceeds as follows:

Algorithm.

1. **Estimate CPs.** Run (1) per occupation o, child age j, and period τ, yielding $\{\alpha_{o,\tau}^{g,j}\}$.

2. **Residualize.** In pre-periods ($\tau \leq \tau_0$), remove occupation and period fixed effects for each gender and child-age group: $\alpha_{o,\tau}^{g,j,\text{res}} = \alpha_{o,\tau}^{g,j} - \bar{\alpha}_o^{g,j} - \bar{\alpha}_\tau^{g,j} + \bar{\alpha}^{g,j}$.

3. **SC weights.** For each occupation and gender, find non-negative weights summing to one that minimize:
$$\min_{w_o^g} \sum_{\tau \leq \tau_0} \sum_j \sum_{y \in \{\text{emp,hrs}\}} \left(\alpha_{o,\tau}^{y,g,j,\text{res}} - \sum_{o'} w_{o,o'}^g \, \alpha_{o',\tau}^{y,g,j,\text{res}} \right)^2, \tag{4}$$
where $w_{o,o'}^g$ is the weight that occupation o' receives as a synthetic control for occupation o, with $w_{o,o}^g = 0$ (each occupation is excluded from its own donor pool), and the summation over y enforces joint matching across employment and hours outcomes.

4. **Debias.** Using the same weights, construct debiased versions of both the static child penalty (from the binary parenthood indicator) and the dynamic child penalties (from each child-age

group): $\tilde{\alpha}^g_{o,\tau} = \alpha^g_{o,\tau} - \sum_{o'} w^g_{o,o'} \alpha^g_{o',\tau}$ and $\tilde{\alpha}^{g,j}_{o,\tau} = \alpha^{g,j}_{o,\tau} - \sum_{o'} w^g_{o,o'} \alpha^{g,j}_{o',\tau}$. Using a single set of weights for both static and dynamic effects ensures a consistent comparison, although results are similar when we compute separate weights. We also debias the treatment: $\tilde{R}^g_o = \Delta R_o - \sum_{o'} w^g_{o,o'} \Delta R_{o'}$. This step is non-standard: in the classic synthetic control framework, donor units are never treated, so $\tilde{R}^g_o = \Delta R_o$. Our generalization allows all occupations to serve as donors even when all are treated to varying degrees.[8]

5. **Estimate.** Regress the post-minus-pre change in debiased CPs on the debiased treatment, using only the post period:

$$\begin{aligned} \text{static:} \quad & \Delta\tilde{\alpha}^g_o = \beta^g \tilde{R}^g_o + \epsilon^g_o, \\ \text{dynamic:} \quad & \Delta\tilde{\alpha}^{g,j}_o = \beta^{g,j} \tilde{R}^g_o + \epsilon^{g,j}_o. \end{aligned} \tag{5}$$

In practice, the static estimand $\Delta\tilde{\alpha}^g_o$ is computed as the equally weighted average of child-age-bin-specific debiased changes: $\Delta\tilde{\alpha}^g_o = \frac{1}{J}\sum_j [(\tilde{\alpha}^{g,\text{post}}_{o,j} - \tilde{\alpha}^{g,\text{pre}}_{o,j})]$.

Three notes are in order. First, for inference, we use randomization inference by permuting the treatment ΔR_o across occupations 1,000 times, keeping SC weights fixed.[9] We compute two-sided p-values as the fraction of permuted test statistics that exceed the actual estimate in absolute value. For confidence intervals, we invert the p-value under a normal approximation.

Second, we also show robustness to using the dummy remote indicator in addition to the continuous treatment version. For the dummy specification, donors for each remote occupation are drawn from non-remote occupations, and vice versa. Third, in the last step of the algorithm where we regress debiased child penalties on debiased exposure, we weight the regression by occupation size to prevent bias from imprecise child penalties of small occupations. We also show robustness to using inverse-variance weights; results remain robust.

[8] The debiased treatment retains substantial variation: for women, $\text{SD}(\tilde{R}^g_o) = 6.7$ compared to $\text{SD}(\Delta R_o) = 6.9$, with a correlation of 0.93 between the raw and debiased treatment. Appendix Figure A.7 shows the scatter.

[9] Fixing SC weights during randomization inference is standard: the weights are functions of the outcome, not the treatment assignment. Re-computing weights for each permutation would conflate the SC step with the inference step.

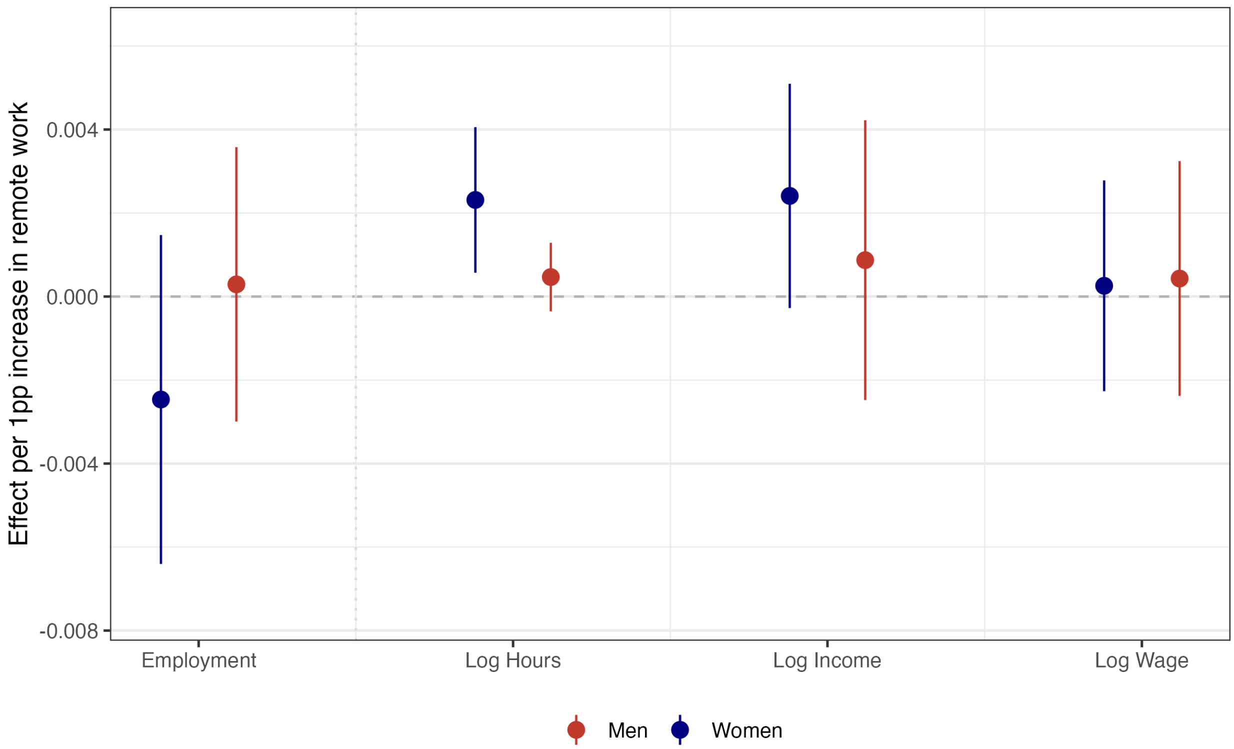
4 Results

Figure 3 reports SC-DiD estimates for four outcomes: employment, and the natural logarithms of hours worked, income, and hourly wages. Our preferred specification uses four child-age bins $j \in \{0,\ 1\text{–}2,\ 3\text{–}5,\ 6\text{–}10\}$, joint matching across employment and hours, and size weights; we assess robustness across 12 specifications that vary the number of child-age bins, the matching moments (separate weights for each outcome as opposed to using the same weights that are estimated to match the trends in employment and hours penalties), and the weighting scheme (occupation size or inverse variance of the occupation-specific child penalties). These robustness checks are reported in Appendix Figure A.2, and we summarize the results here. A positive coefficient indicates that the child penalty is becoming less negative, that is, the penalty is shrinking.

Our first result is that remote work does not reduce the employment child penalty for mothers. The point estimate in our preferred specification is negative (-0.0025, $p = 0.303$), indicating that if anything the employment penalty increases in occupations that adopted more remote work. This finding contrasts with the positive correlation between remote work and mothers' employment documented by Harrington and Kahn (2026) using pre-COVID data. As we show in Section 3.2, remote occupations were already on a differential downward trajectory in employment penalties throughout 2001–2019. Our synthetic control adjustment removes this pre-existing trend, and the apparent positive association disappears. We return to the suggestive negative employment effect in Section 4.3, where we discuss evidence that remote work may lower hourly productivity for some mothers.

Second, we find robust evidence that remote work increases mothers' hours conditional on employment. In our preferred specification, each percentage point increase in an occupation's remote work share raises log hours by 0.23% ($p = 0.029$). To understand the economic significance of these effects, note that the average remote occupation experienced an 8.9 pp larger increase in remote work share relative to the average non-remote occupation. The implied effect is $8.9 \times 0.23\% \approx 2\%$ more hours worked. At baseline (2016–19), mothers in remote occupations worked 12.6% fewer hours than their pre-birth counterfactual, so this effect eliminates roughly one-sixth of the hours child penalty. For comparison, Ciasullo and Uccioli (2023) find that Australia's Fair Work Act, which entitled mothers to request reduced hours on permanent contracts, reduced the hours penalty by

Figure 3: SC-DiD: Effect of Remote Work on Child Penalties



Notes: Each panel reports the SC-DiD treatment effect of occupation-level remote work adoption (ΔR_o, continuous, per 1 pp) on the child penalty for a given outcome. Blue markers denote women; red markers denote men. Bars show 95% confidence intervals from randomization inference (1,000 permutations). *Step 1.* For each occupation o, gender g, time period τ (2001–05, 2006–10, 2011–15, 2016–19, 2022–24), and child-age bin j (= 0, 1–2, 3–5, 6–10), we estimate the child penalty $\alpha_{o,j}^{g,\tau}$ from a regression of the outcome on parenthood indicators, controlling for age, calendar year, and education fixed effects. Employment penalties are estimated on the full sample interacting the dependent variable with an occupation indicator; conditional outcomes (hours, income, wages) are estimated within occupation among the employed. Employment effects are normalized by the counterfactual (predicted outcome absent the penalty). *Step 2.* For each occupation, we construct a synthetic control as a convex combination of all other occupations, with weights chosen to match pre-period residualized child penalties jointly across employment and hours and across all four child-age bins. We then compute the SC-debiased DiD estimand: $\widetilde{\Delta\alpha}_o^g = \frac{1}{J}\sum_j\big[(\widetilde{\alpha}_{o,j}^{g,\mathrm{post}} - \widetilde{\alpha}_{o,j}^{g,\mathrm{pre}})\big]$, where $\widetilde{\alpha}$ denotes the SC-residualized penalty. The reported coefficient β is the slope from regressing $\widetilde{\Delta\alpha}_o^g$ on the SC-debiased treatment $\tilde{R}_o^g$, weighted by occupation size.

22%. A market-driven change in work arrangements that eliminates 16% of the hours penalty without government intervention is economically meaningful, particularly given that traditional family policies have had limited effects on child penalties (Kleven et al., 2024b).

We find no effect of remote work on mothers' income conditional on employment. The estimate is not statistically significant in the preferred specification, and estimates are near zero and mixed in sign across the robustness checks, with none reaching statistical significance (smallest $p = 0.140$).

Remote work increases hours but does not increase earnings, consistent with the additional hours being less productive or with mothers accepting lower wages in exchange for the flexibility of remote work. Interestingly, we also find no effect on hourly wages: point estimates are near zero across robustness checks, mixed in sign, and all statistically insignificant (smallest $p = 0.115$). The additional hours worked by mothers in remote occupations do not translate into higher hourly compensation.

For men, we find no average effects on any outcome. Point estimates are close to zero and statistically insignificant across all four outcomes. The contrast between genders is consistent with remote work operating through flexibility for the parent who bears the disproportionate share of caregiving, rather than through occupation-level demand shocks, which would affect both genders similarly.

4.1 Robustness

We assess robustness along several dimensions. First, we vary the specification: across 12 combinations of child-age bins, matching moments, and weighting schemes, the hours effect is positive in all 12, statistically significant at the 5% level in five, and at the 10% level in nine (Appendix Figure A.2). The employment effect is negative in all 12 specifications and marginally significant in one ($p = 0.055$). Income and wage effects are null throughout. These patterns confirm that the hours result is not an artifact of specification choice, and that the employment, income, and wage nulls are robust.

Backtest. We backtest the synthetic control weights following the best practice recommended by Abadie (2021). We restrict the training period to 2001–2015, assign a placebo treatment to 2016–2019, and test for spurious effects. All placebo tests for women yield precise nulls; the smallest p-value for women's log hours is 0.316. This suggests that the SC weights capture genuine treatment-period signal rather than overfitting to pre-period noise, though the test cannot rule out confounding from shocks concurrent with the pandemic.

Dummy treatment. Our main specification treats remote work intensity as continuous, exploiting variation in the magnitude of ΔR_o across occupations. An alternative is to dichotomize: classify occupations as "remote" or "non-remote" based on whether ΔR_o exceeds a threshold, and apply the

Synthetic Difference-in-Differences (SDiD) estimator of Arkhangelsky et al. (2021). This approach requires two researcher choices that our continuous specification avoids: the percentile threshold that defines treatment, and the number of child-age bins over which to average occupation-level penalties before applying SDiD. Appendix Figure A.3 reports Wald-scaled estimates for women across all 18 combinations of these choices (6 thresholds from the 30th to the 80th percentile, and 3 child-age bin configurations). The hours effect is positive in nearly all specifications, consistent with our main finding, though imprecise due to the loss of information from dichotomizing a continuous treatment. Income and wage effects are null throughout. The employment effect is sensitive to the threshold choice, turning positive at higher percentiles where fewer, more intensely treated occupations are classified as remote.

CPS replication. We replicate the analysis using the CPS rotating panel instead of the ACS pseudo-panel, following the rotating-panel approach of Gulek (2024). The rotating panel method does not rely on matching, but the CPS contains roughly nine times fewer parents with newborns per year. The CPS estimate is too imprecise to independently confirm the hours effect, but it is consistent with the ACS result in sign and magnitude (Appendix Figure A.4).

Anticipatory sorting. We test whether individuals anticipatorily sort into remote occupations before becoming parents. Using the pseudo-panel matching, we assign event time $t = -2$ to non-parents observed two years before their matched parent's first child, in addition to the existing $t = -1$ assignment. We then estimate an event study of the probability of being employed in a remote occupation, interacting the $t = -1$ indicator with period dummies. If future parents differentially enter remote occupations in anticipation of parenthood after COVID, the post-period coefficient should be positive. Appendix Figure A.9 shows that the 2022–24 coefficient is 0.002 for women ($p = 0.762$) and 0.008 for men ($p = 0.217$); all coefficients across all periods are statistically insignificant for both genders. We find no evidence of anticipatory sorting.

Heterogeneity. We explore heterogeneity along two dimensions (Appendix Figure A.8). The hours effect is positive for both white (+0.0023, $p = 0.022$) and non-white (+0.0017, $p = 0.251$) women; the two estimates are not statistically distinguishable, and the difference in significance likely reflects the smaller non-white subsample. The effect is larger for young parents aged 25–34

(+0.0027, $p = 0.061$) than for older parents aged 35–45 (+0.0012, $p = 0.290$), though the difference between the two estimates is not statistically significant.

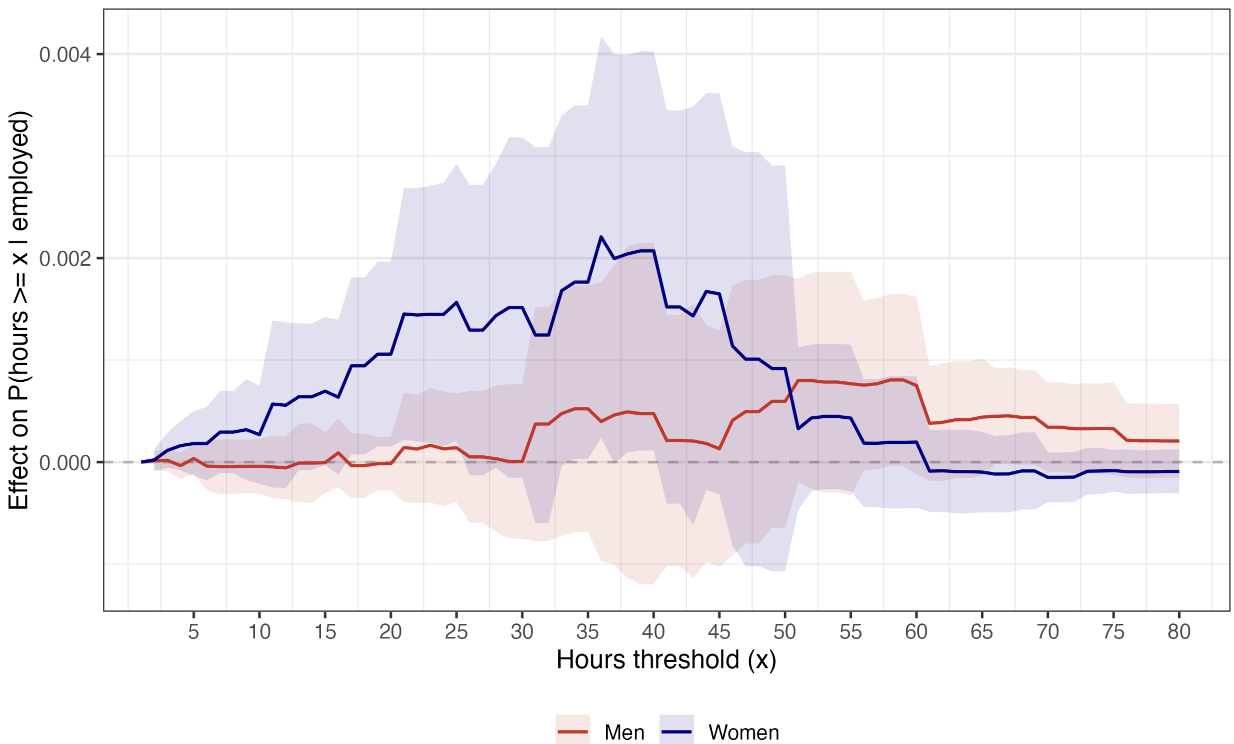
4.2 Distributional effects on hours

Given that a decrease in hours penalty is the only robust effect of remote work on child penalties, we investigate where in the hours distribution this effect originates. How does remote work impact the hours penalty across the hours distribution? Does remote work shift mothers from very low hours to moderate hours, or from moderate to full-time? To answer these questions, for each threshold $x \in \{2, 3, \ldots, 80\}$, we define $Y_x = \mathbf{1}\{\text{hours} \geq x\}$ conditional on employment and apply the SC-DiD procedure to these binary outcomes. The child penalty with this outcome measures: conditional on working, how does parenthood impact the probability of working more than x hours? We analyze how remote work impacts this penalty.

Note that the treatment effect is mechanically zero at the boundaries: conditional on employment, virtually everyone works at least one hour and almost no one works more than 80 hours, so remote work cannot change these probabilities. Between these boundaries, the effect can be positive or negative at each threshold. A positive effect at threshold x means that remote work increases the probability that mothers work at least x hours per week; equivalently, it shifts mass from below x to above x.

Figure 4 reports the treatment effect at each threshold for both genders. For women, the effect is near zero below around 13 hours per week, becomes positive and statistically significant around 17 hours, and remains elevated through 45 hours, with a peak at 36 hours (+0.0022, $p = 0.028$). The effect is statistically significant at the 5% level for thresholds between 17 and 25 hours and between 34 and 40 hours. Above 45 hours, the effect gradually declines and returns to zero. This pattern indicates that remote work shifts mothers from low part-time hours (below approximately 15 hours per week) toward substantial part-time or full-time hours (25–45 hours per week). The effect is not concentrated at any single margin; instead, it operates broadly across the middle of the distribution, consistent with the baseline hours penalty being largest at the full-time threshold of 35–40 hours (Appendix Figure A.5). Because the pseudo-panel does not track individuals, we cannot rule out that the distributional shift partly reflects changes in the composition of employed mothers rather than individual behavioral changes, though the null employment effect limits the scope for such compositional shifts.

Figure 4: Distributional Effects of Remote Work on Hours



Notes: SC-DiD treatment effect of occupation-level remote work adoption (ΔR_o, continuous, per 1 pp) on Pr(hours $\geq x$ | employed) at each threshold $x = 2, 3, \ldots, 80$. Blue: women. Red: men. SC weights are the same as in Figure 3 (joint matching across employment and hours, four child-age bins, size-weighted). Shaded bands show 95% confidence intervals from randomization inference (1,000 permutations). The effect is mechanically zero at the boundaries. Pointwise p-values are unadjusted; the figure characterizes the shape of the distributional effect rather than providing 79 independent hypothesis tests.

For men, the treatment effect is near zero throughout the lower and middle portions of the distribution. A suggestive but statistically insignificant positive pattern emerges in the right tail (above 50 hours per week), though the estimates are noisy and do not reach conventional significance levels.

4.3 Secondary childcare during work

The finding that mothers in remote occupations work more hours without earning more suggests that remote working hours may become less productive. One explanation is that remote work allows mothers to supervise children while working, lowering hourly productivity. We test this channel using the ATUS, which records secondary childcare: minutes during which a respondent supervises a child while engaged in another activity. We focus on secondary childcare that occurs during paid work.

We collapse individual-level ATUS data to four cells (remote × period) using survey weights and estimate a difference-in-differences comparing remote and non-remote occupations before (2016–19) and after (2022–24) COVID. We use a simple difference-in-differences rather than the SC-DiD applied to the main results because pre-trends in secondary childcare are parallel for 2011–2019, which justifies the simpler design. For inference, we permute the remote label across occupations (1,000 draws).[10]

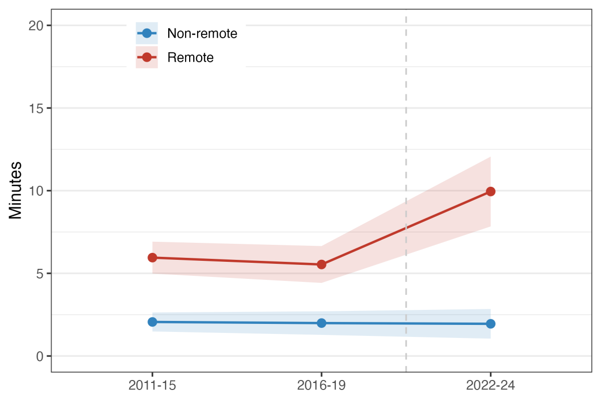
Figure 5 presents the results. Panels (a) and (b) show mean secondary childcare during work by period, separately for fathers and mothers. In both panels, remote and non-remote occupations follow nearly identical trends through 2011–19, then diverge sharply in 2022–24. The divergence is larger for mothers: secondary childcare during work in remote occupations rises from 12 to 31 minutes per day for mothers, compared to 13 to 23 minutes for fathers, while non-remote occupations changed from 10.5 to 14.7 minutes for mothers and remained stable at 4.4 minutes for fathers. The difference-in-differences estimate is 15.5 minutes per day for mothers ($p = 0.005$) and 10.7 minutes for fathers ($p = 0.004$), relative to pre-treatment means of 11.6 minutes for mothers and 12.5 minutes for fathers, implying a 133% increase for mothers and an 86% increase for fathers.[11]

Panel (c) shows where in the distribution this effect originates. We collapse individual-level ATUS data to four cells (remote × period) using survey weights and estimate the difference-in-differences at each percentile. Below the 87th percentile, the treatment effect is zero for both genders: most parents in remote occupations do no secondary childcare during work, as was the case before COVID. The entire mean effect is driven by the upper tail. At the 91st percentile, mothers' treatment effect reaches 100 minutes ($p = 0.008$) and fathers' reaches 25 minutes ($p = 0.001$). Mothers' effect rises earlier and more steeply than fathers', diverging from zero around the 87th percentile compared to the 91st for fathers, consistent with mothers bearing a disproportionate

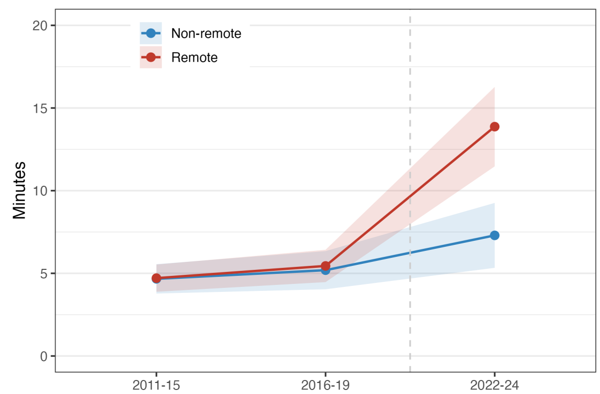
[10] Periods before 2011 are excluded because ATUS redesigned the secondary childcare module in 2010, creating a measurement break. The three periods used (2011–15, 2016–19, and 2022–24) provide one pre-trend check and one treatment comparison. Placebo tests comparing 2011–15 to 2016–19 yield near-zero estimates for both genders.

[11] Fathers' pre-treatment levels differ substantially between remote (12.5 min) and non-remote (4.4 min) occupations. A common proportional increase in childcare during work would generate a larger absolute DiD for the remote group even absent a treatment effect. This concern is mitigated for mothers, where pre-treatment means are closer (11.6 vs. 10.5 min).

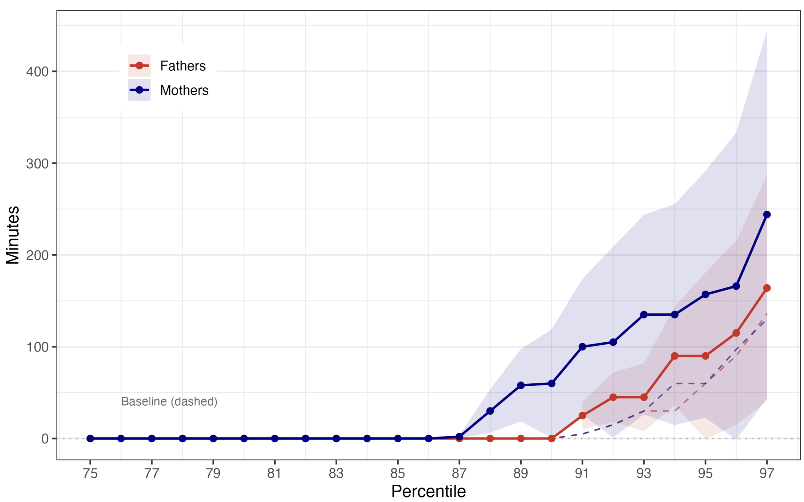
Figure 5: Secondary Childcare During Paid Work (ATUS)



(a) Fathers



(b) Mothers



(c) Quantile treatment effects

Notes: Panels (a)–(b): Mean secondary childcare minutes during paid work, by period and remote status. ATUS, employed parents aged 18–64, survey-weighted, 95% CIs. Remote and non-remote defined by median split of ΔR_o (same classification as main analysis). Periods before 2011 excluded due to ATUS measurement redesign. Panel (c): Difference-in-differences at each percentile (2016–19 vs. 2022–24). Individual-level data collapsed to four cells (remote $\times$ period) using survey weights. Solid lines show DiD estimates; dashed lines show pre-treatment baseline levels in remote occupations. 95% CIs from randomization inference (1,000 permutations of the occupation-level remote label).

share of childcare responsibilities when working from home.

These results are consistent with the productivity channel: a subset of mothers in remote occupations are working while simultaneously supervising children, with mothers above the 90th percentile of secondary childcare averaging more than 60 minutes per day of concurrent childcare and paid work. Working while supervising children may lower hourly productivity, which explains why additional hours have not translated into higher income. The finding that fathers also increase childcare during work, but less than mothers, is consistent with remote work shifting some caregiving to fathers without fully equalizing the division of childcare.

5 Conclusion

Remote work reduces the hours child penalty for mothers but does not affect employment, income, or wage penalties. For the average remote occupation, the implied effect is approximately 2% more hours worked, eliminating one-sixth of the baseline hours penalty. We find no average effects on men. Results remain robust across several specification, weighting, and matching choices. ATUS evidence suggests that the absence of income gains may reflect lower productivity: mothers in remote occupations sharply increased the time spent supervising children while working after COVID.

Occupations that eventually adopted remote work were already on differential child penalty trajectories before COVID, and any causal analysis of remote work and child penalties must account for these pre-existing trends. We address this using a synthetic control estimator that generalizes to continuous treatments (Gulek and Vives-i Bastida, 2024). To estimate occupation-level child penalties with sufficient precision, we adapt the pseudo-panel approach of Kleven (2025) to the ACS while excluding the 2020 and 2021 survey years, avoiding potential structural breaks induced by the pandemic. This combination of pseudo-panels and SC-DiD is generalizable: it can be employed in any setting where large cross-sectional data exist, and the Child Penalty Atlas documents child penalties across 134 countries where this approach could be applied (Kleven et al., 2024a).

Remote work did not decrease mothers' employment penalties, but employment penalties in eventually-remote occupations did decrease throughout the 2000s and 2010s, well before COVID. Possible drivers include improvements in communication technology, online conferencing, and changing workplace norms, though the underlying mechanisms remain unidentified. Identifying which changes drove this secular convergence, whether similar changes can be fostered in occupations where remote work is not feasible, and what policy levers can accelerate them, are important questions for future research.

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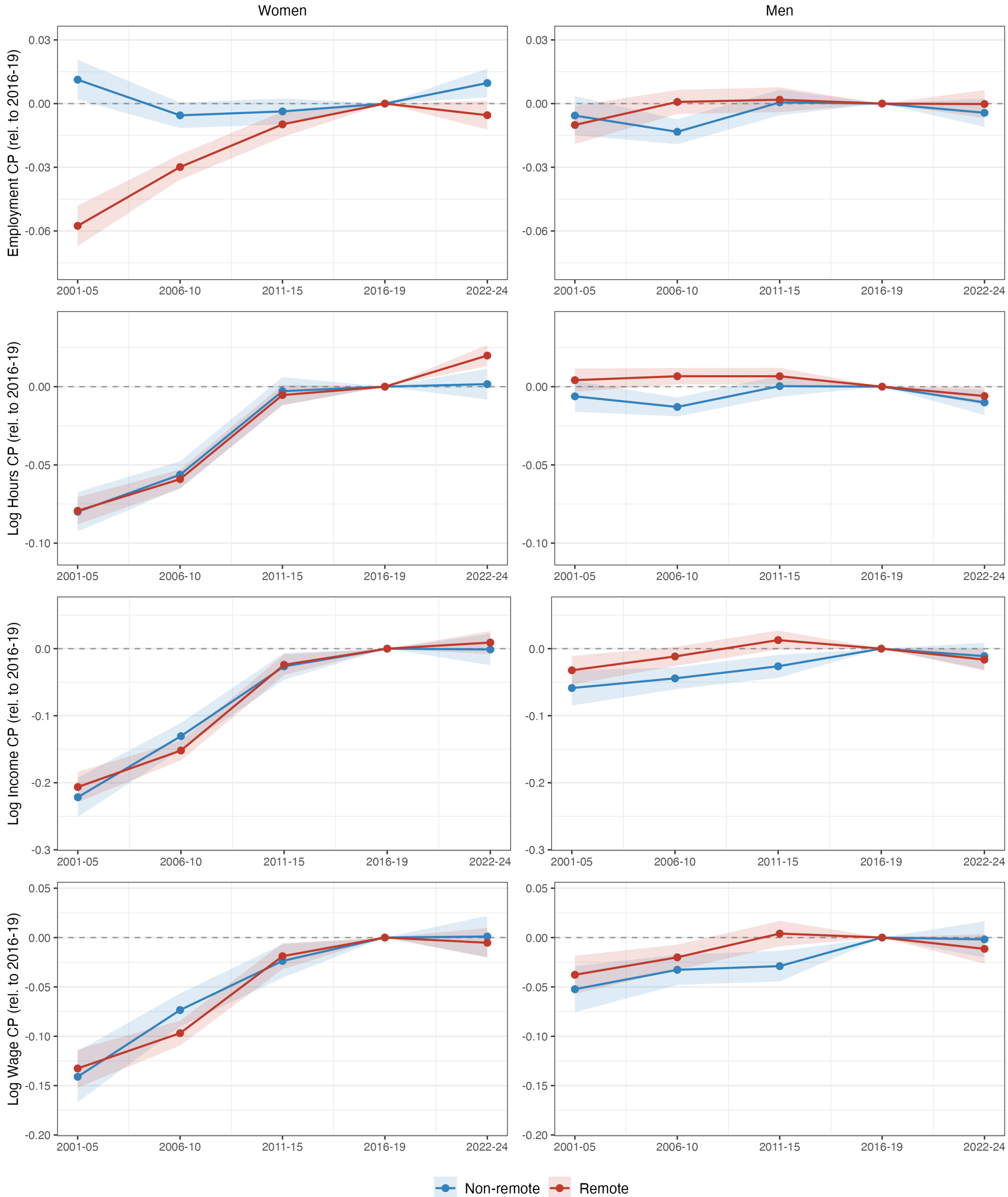
A Online Appendix

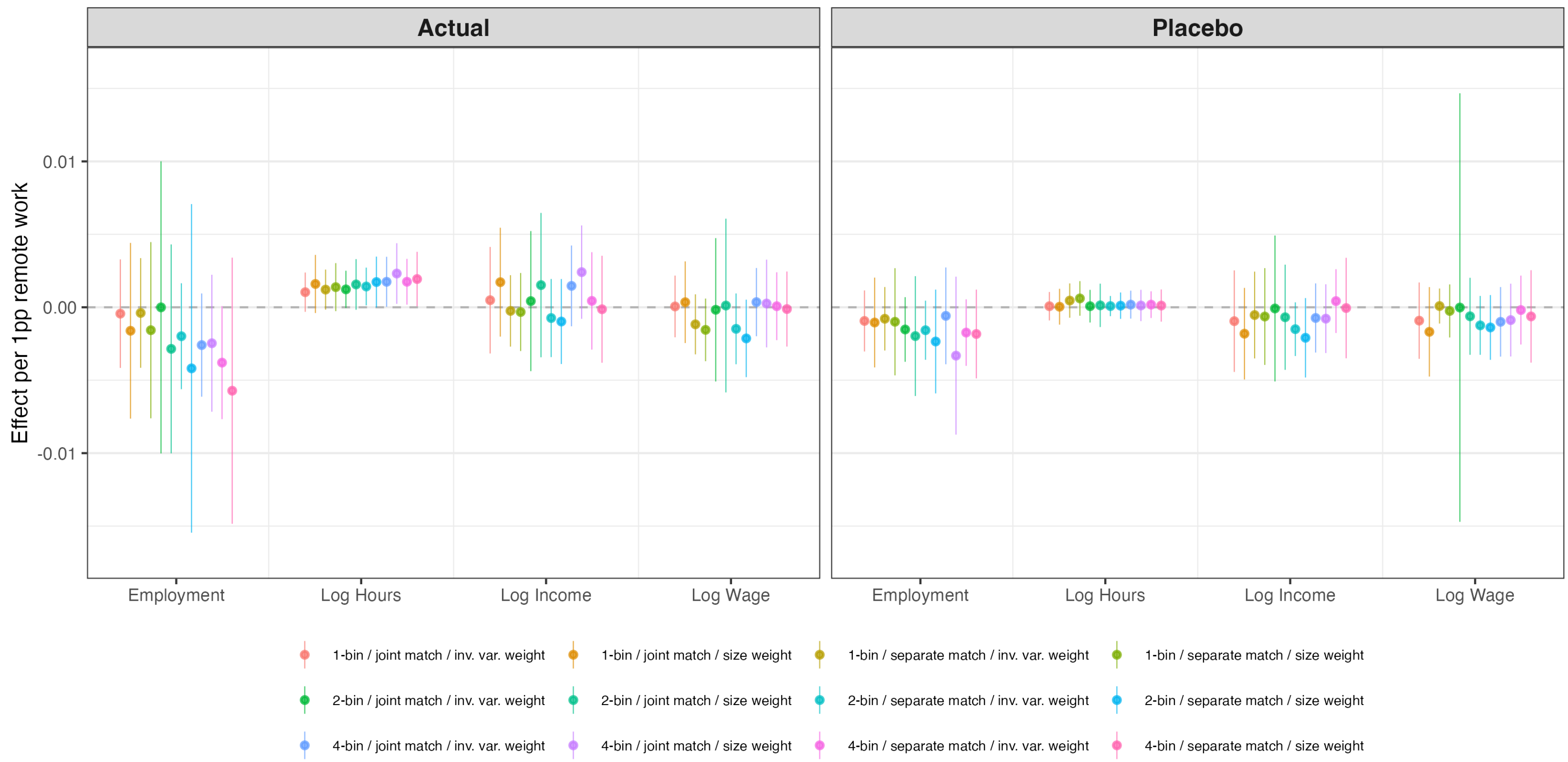
Contents

Figure A.1: Child Penalty Trajectories by Outcome (ACS, Both Genders)



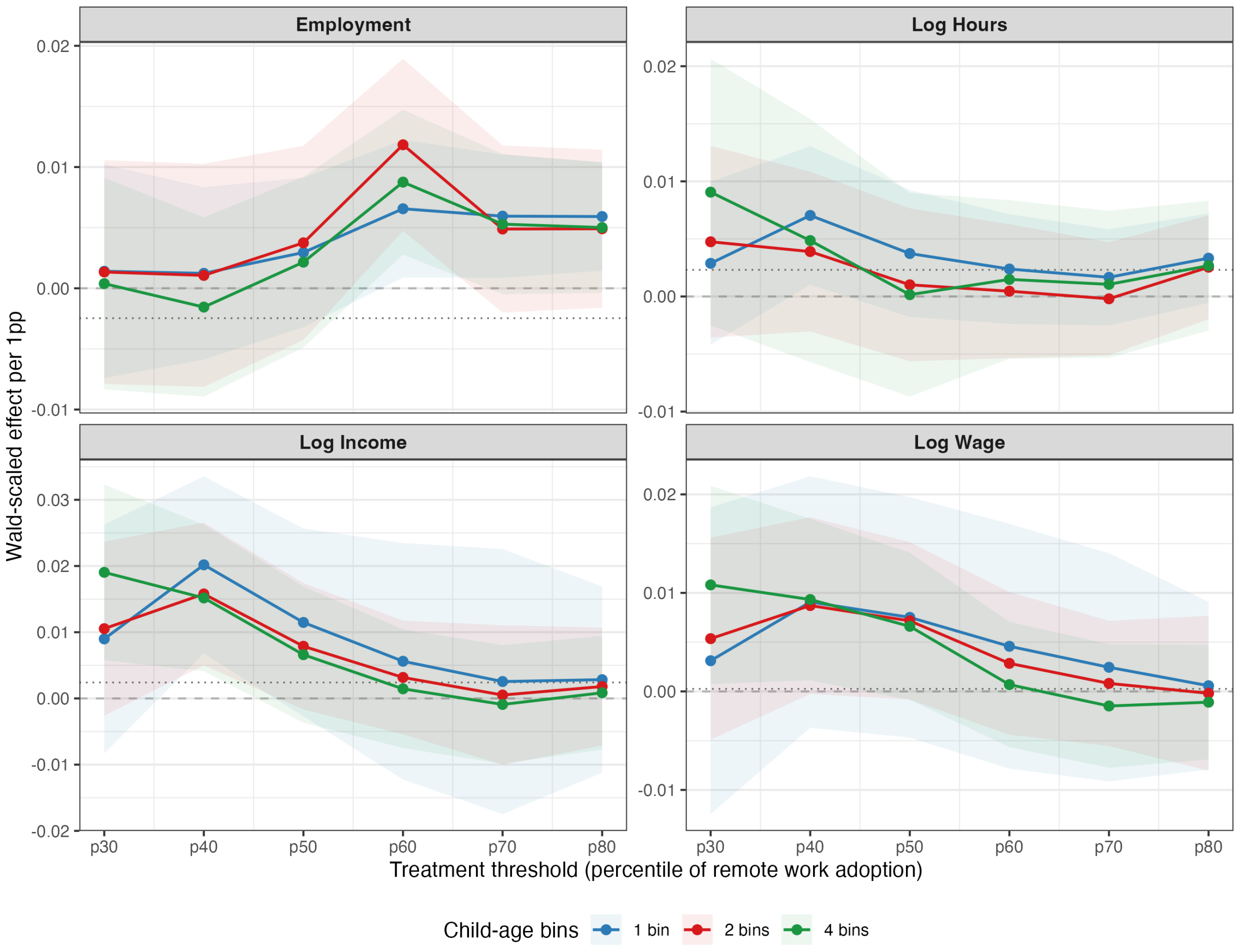
Notes: Child penalty deviations relative to the reference period (2016–19, normalized to zero). Left column: women. Right column: men. Red: remote occupations ($\Delta R_o \geq$ median). Blue: non-remote. ACS pseudo-panel, five periods: 2001–05, 2006–10, 2011–15, 2016–19 (reference), 2022–24. Y-axes are shared within each row for comparability across genders. Women's employment penalties show differential pre-trends across remote and non-remote occupations; conditional outcomes (log hours, log income, log wages) show parallel pre-trends, validating the parallel trends assumption for these outcomes.

Figure A.2: Robustness Across Specifications: Actual vs. Placebo (Women)



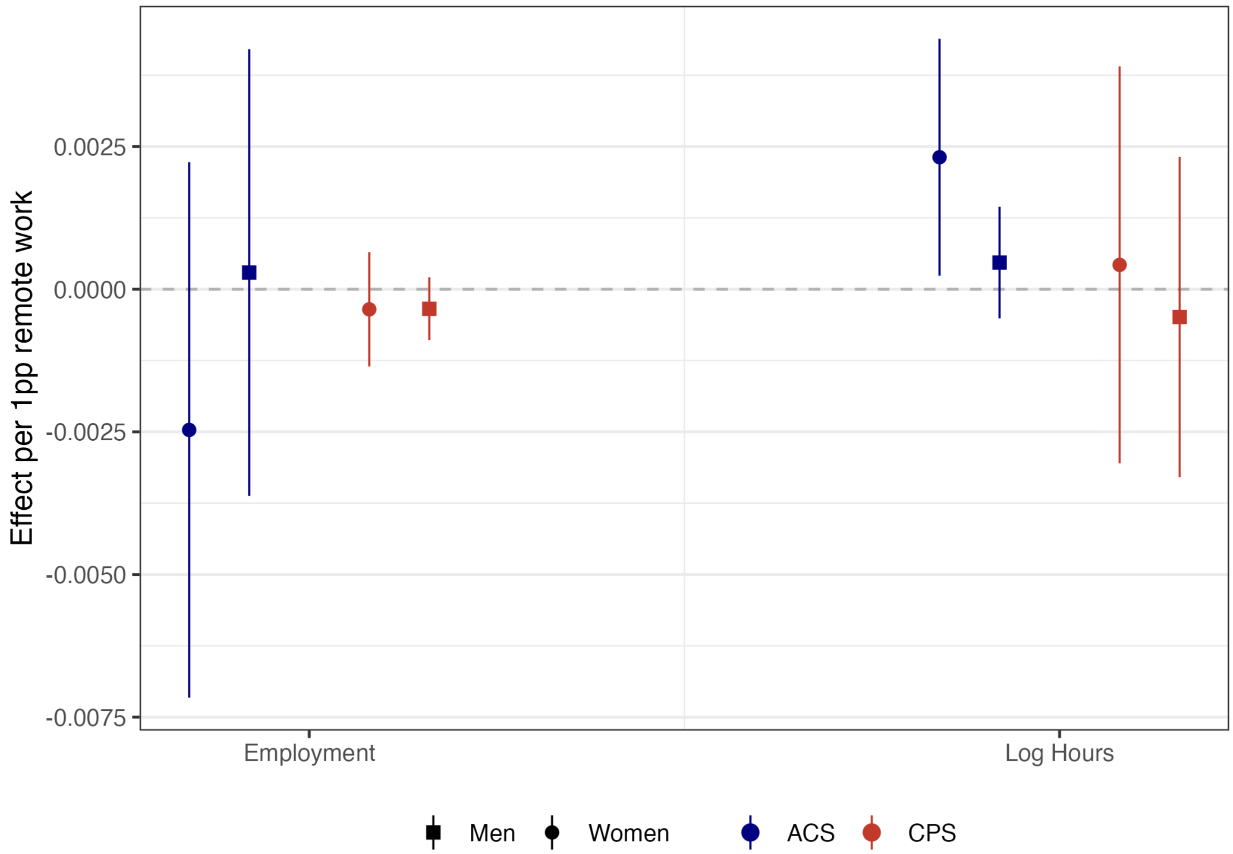
Notes: SC-DiD estimates across 12 specifications: 3 child-age bin configurations (1, 2, or 4 bins) $\times$ 2 matching approaches (separate weights per outcome vs. joint weights across employment and hours) $\times$ 2 weighting schemes (occupation size vs. inverse variance). Each point is one specification. Continuous treatment ΔR_o. 95% CIs from randomization inference (1,000 permutations). Left panel: actual estimates (2016–19 vs. 2022–24). Right panel: placebo backtest (training on 2001–15, testing at 2016–19). The hours effect is positive in all 12 actual specifications and zero in all 12 placebo specifications. The employment effect is negative in all 12 actual specifications; placebo employment estimates are near zero. Income and wages are null in both panels. For men, all actual and placebo estimates are close to zero across all four outcomes and all 12 specifications.

Figure A.3: SDiD Robustness: Across Thresholds and Child-Age Bins (Women)



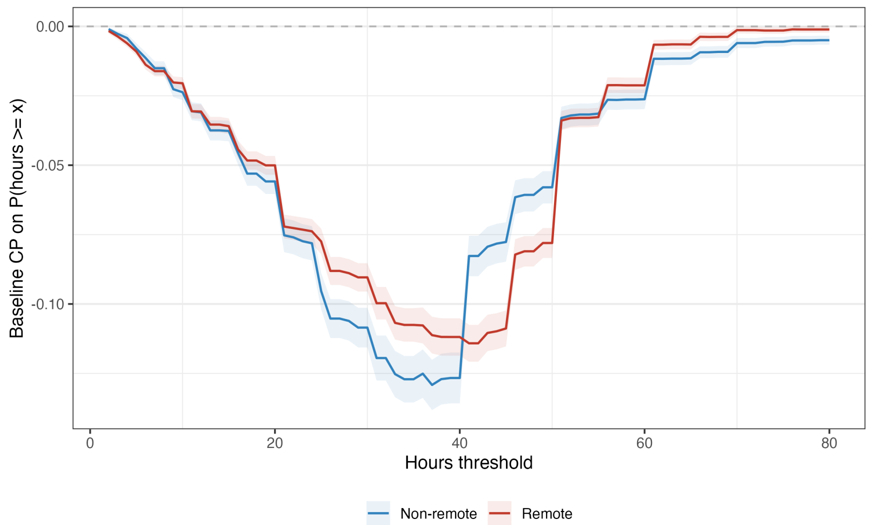
Notes: Synthetic Difference-in-Differences (SDiD) estimates of Arkhangelsky et al. (2021) for women, using a binary remote/non-remote classification. Each panel shows one outcome. The x-axis varies the percentile threshold that defines treatment: at the p-th percentile, occupations with ΔR_o above the p-th percentile are classified as treated. Three child-age bin specifications are shown (1, 2, or 4 bins). Coefficients are Wald-scaled by the mean gap in ΔR_o between treated and control groups at each threshold, making magnitudes comparable to the continuous SC-DiD in Figure 3. Dotted horizontal line: main SC-DiD continuous estimate ($\hat{\beta}$, es4jbin, joint, size-weighted). 90% CIs from randomization inference (1,000 permutations). The hours effect is positive across nearly all 18 specifications, consistent with the main finding. Employment is sensitive to the threshold choice. Income and wages are null throughout. All SDiD estimates for men are null (not shown).

Figure A.4: Robustness: Pseudo-Panel (ACS) vs. Rotating-Panel (CPS)



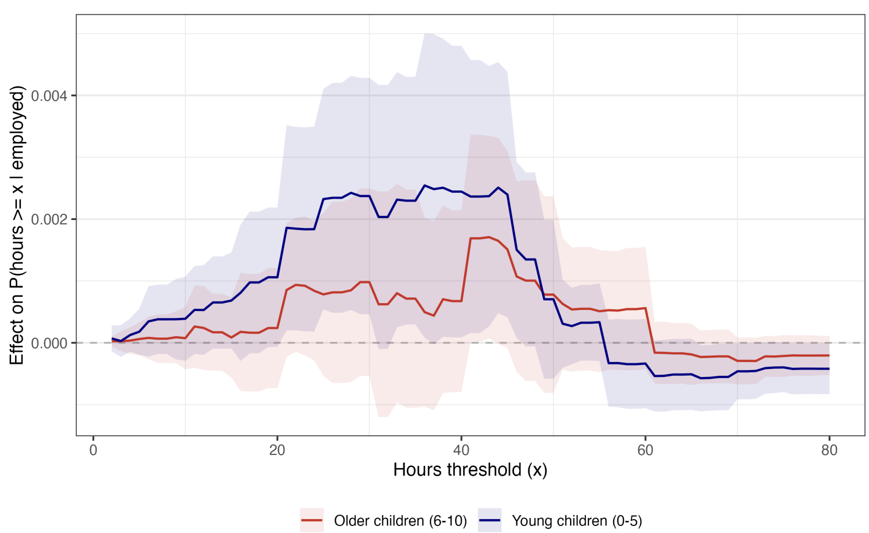
Notes: SC-DiD estimates comparing the pseudo-panel approach (ACS, blue) with the rotating-panel approach (CPS, red). The ACS pseudo-panel matches parents to observationally similar non-parents using age, education, race, and state; the CPS rotating panel directly observes individuals before and after their first child's birth without matching. Circles: women. Squares: men. Continuous treatment ΔR_o, four child-age bins, size-weighted. RI 95% CIs from 1,000 permutations. The CPS estimate is consistent with the ACS in sign and magnitude, though too imprecise to independently confirm the effect. The CPS contains $\sim 9\times$ fewer parents with newborns per year than the ACS, explaining the wider confidence intervals.

Figure A.5: Baseline Distributional Hours Penalties



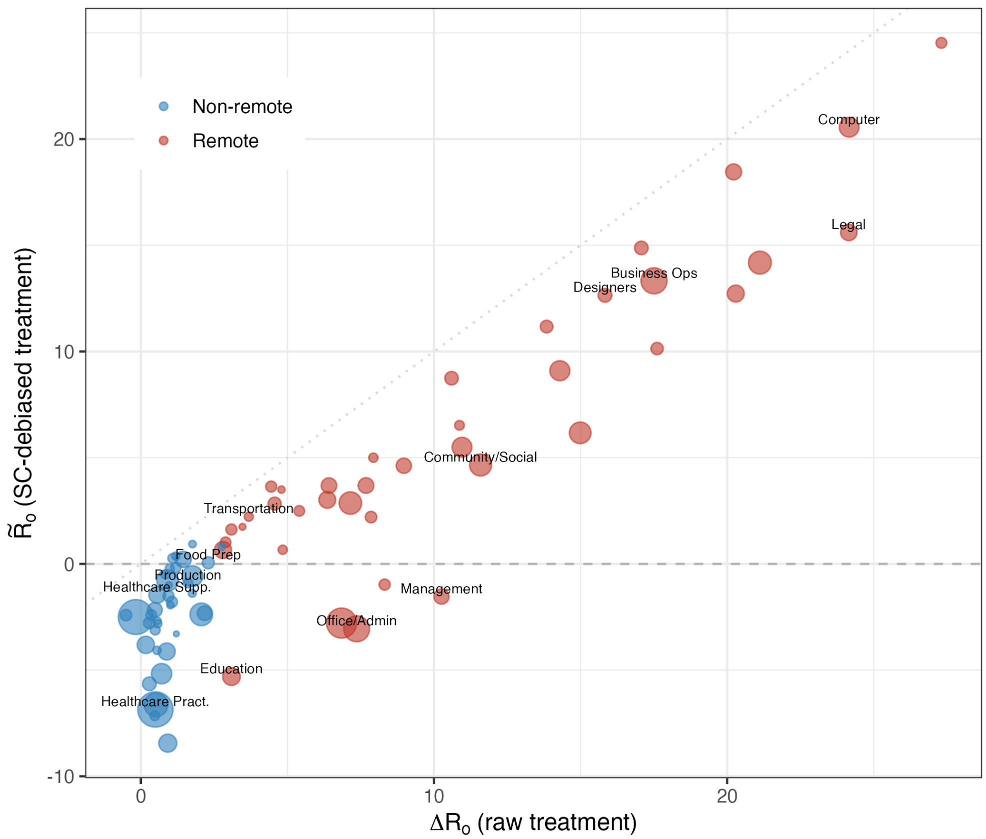
Notes: Baseline child penalty on Pr(hours $\geq x$ | employed) at each threshold x, estimated in the reference period (2016–19). Red: remote occupations. Blue: non-remote. The penalty measures how much motherhood reduces the probability of working at least x hours per week. Women's penalty is concentrated at $x = 35$–40 hours (the full-time threshold), with non-remote occupations exhibiting larger penalties than remote occupations at $x = 35$–45. This context helps interpret Figure 4: the treatment effect of remote work is largest where the baseline penalty is largest.

Figure A.6: Distributional Effects by Child Age (Women)



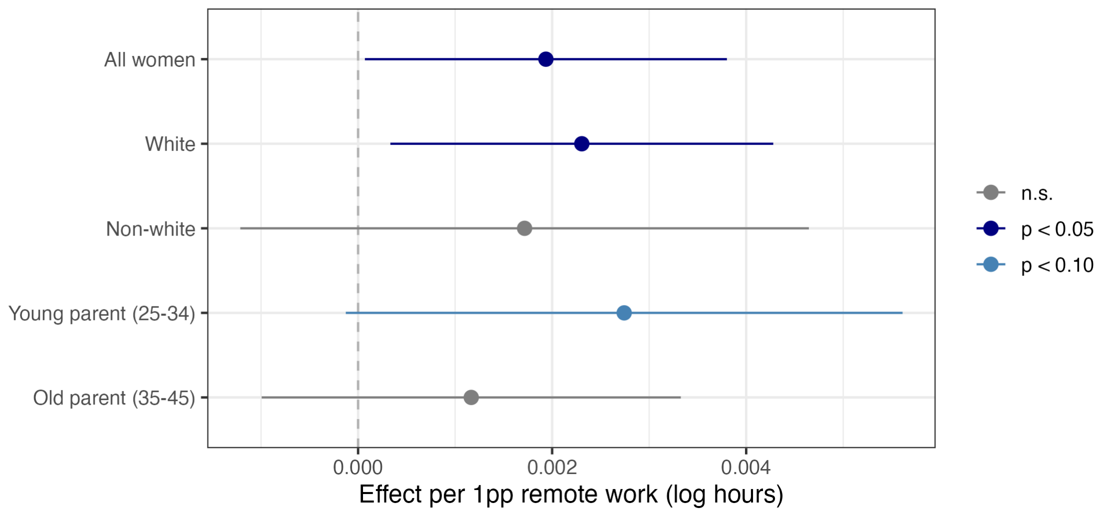
Notes: SC-DiD treatment effect of remote work on Pr(hours $\geq x$ | employed) at each threshold x, estimated separately for mothers with young children (age 0–5) and mothers with older children (age 6–10). Same SC weights as in the main specification (joint matching across employment and hours, four child-age bins, size-weighted). Continuous treatment ΔR_o. RI 95% CIs from 1,000 permutations. The distributional effect is concentrated among mothers with young children, consistent with remote work flexibility being most valuable when caregiving demands are highest.

Figure A.7: Raw vs. SC-Debiased Treatment (Women)



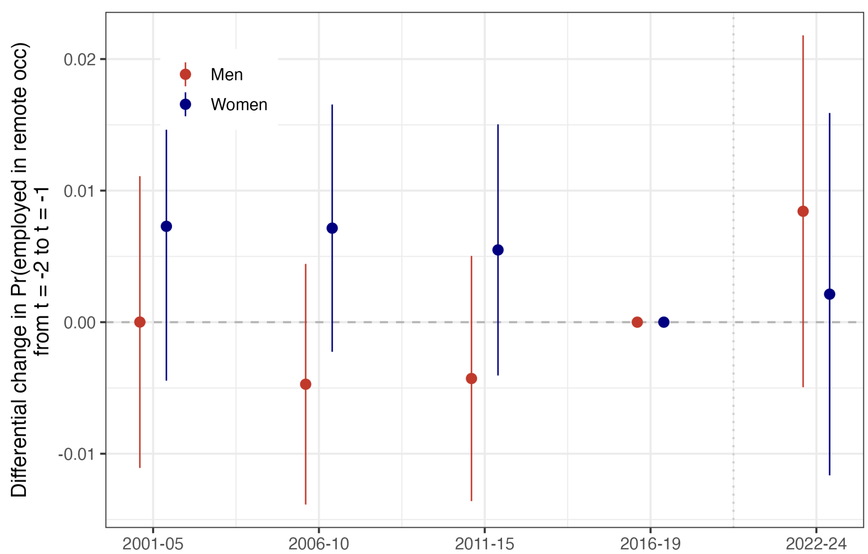
Notes: Each bubble is a 3-digit SOC occupation, sized by employment. x-axis: raw treatment ΔR_o (change in remote work share, pp). y-axis: SC-debiased treatment $\tilde{R}_o^g$ for women. Dotted line: 45-degree line. Dashed line: zero. The correlation between raw and debiased treatment is 0.93, and the variance ratio $\mathrm{Var}(\tilde{R}_o^g)/\mathrm{Var}(\Delta R_o) = 0.93$, indicating that SC debiasing retains 93% of the treatment variation.

Figure A.8: Heterogeneity in the Hours Effect (Women)



Notes: SC-DiD estimates of the effect of remote work on mothers' log hours, by subgroup. Continuous treatment ΔR_o, four child-age bins, separate matching, size-weighted. 95% CIs from randomization inference (1,000 permutations). Young parents are aged 25–34 at first birth; old parents are aged 35–45.

Figure A.9: Anticipatory Occupational Sorting into Remote Occupations



Notes: Each coefficient measures the differential change in Pr(employed in a remote occupation) from event time $t = -2$ to $t = -1$ for matched non-parents, relative to the reference period (2016–19). Non-parents at $t = -1$ are drawn from ACS year $T - 1$ and matched to new parents in year T on sex, education, marital status, race, birth cohort, and state; non-parents at $t = -2$ are drawn from ACS year $T - 2$ using the same matching cell. All ACS years 2020–2021 are excluded. Regression includes age, year, and education fixed effects, weighted by survey weights. 95% CIs from robust standard errors. If individuals anticipatorily sort into remote occupations before having children after COVID, the 2022–24 coefficient should be positive. For women, the post-COVID coefficient is 0.002 ($p = 0.762$); all coefficients are statistically insignificant for both genders.